

Managing Processes



Overview

- User Processes.
- System processes.
- Process utilities.
- Process locks.



User Processes

- When user connects, process started can be interactive or background process.
 - Interactive: user logging in to Terminal session.
 - Process listing displays “shell” or name of running routine in Routine column.
 - Background: code uses ObjectScript JOB command.

Interactive Process Examples	Background Process Examples
▪ Terminal session ▪ Application that runs in Terminal	▪ Jobs for incoming external connections (Java, xDBC) ▪ Jobs for Web Applications (Portal) ▪ Jobs for processing large amounts of data (reports)



User Process in Batch Mode

- Process that uses large number of global buffers can adversely affect other users.
 - Most often background process but can be interactive.
- Set process batch flag to restrict process to 25% of global buffers.
 - Other users share remaining 75% and continue working effectively while batch runs.
- Enable/disable process batch flag in code.

```
do $system.Process.BatchFlag(1)    // enable
do $system.Process.BatchFlag(0)    // disable
```



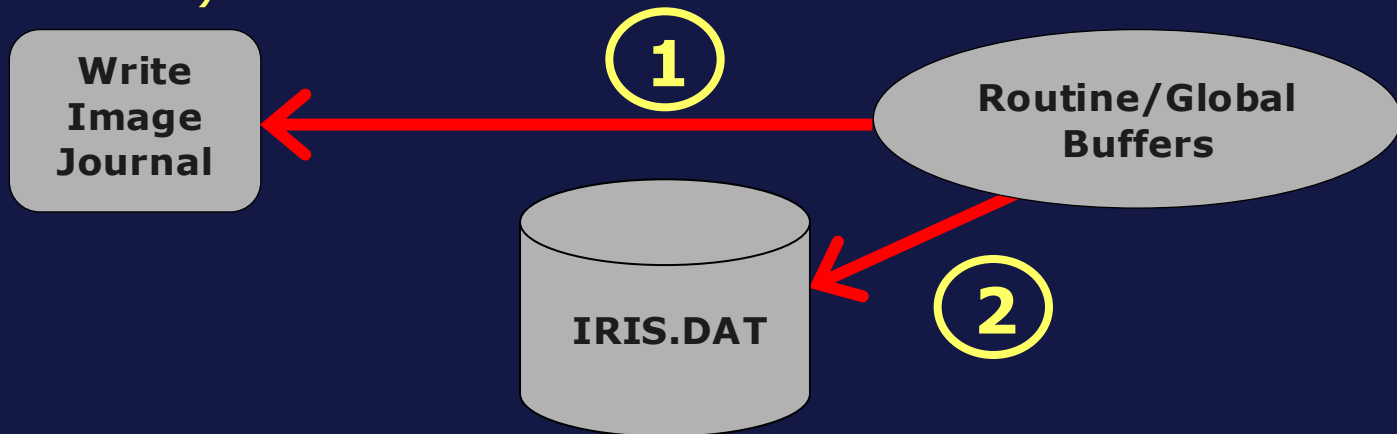
CONTROL

- Starts as first process when system is started.
- Creates Write Daemon, Journal Daemon, Garbage Collector, Expansion Daemon and network processes.
- Notifies processes of pending messages, such as broadcast message.
- Perform other functions on various platforms.



WRTDMN

- Performs all writes to database from buffers.
 - Frequency of Write Daemon pass based on activity.
 - At least every 80 seconds.
- Performs writes into Write Image Journal from buffers.
- Eight daemons (7 are Auxiliary Write Daemons).
 - Windows platforms have one Write Daemon (multi-threaded).



Other System Daemons

- **GARCOL.**
 - Analyzes and frees database space when global is removed.
- **EXPDMN.**
 - Expands database when necessary.
- **JRNDMN.**
 - Writes journal buffer contents to journal file.
- **CLNDMN.**
 - Cleans ghost processes (process exists in IRIS but not at OS level).
 - Removes locks, rolls back transactions, removes entry from processes table and more.



System Processes

- Super Server.
 - Controls connections to system through Super Server port.
 - Process listing displays %SYS.SERVER in Routine column.
- LMFMON.
 - License monitor.
- Task Manager. (Automation module)
 - Process listing displays %SYS.TaskSuper.1 in Routine column.



System Processes (cont.)

- MONITOR.
 - System Monitor. (System Monitoring module)
 - Monitors and reports system errors.
- %SYS.cspServer# processes.
 - Web application requests.
 - Including REST requests.



Process Utilities

- Allows control of processes:
 - System Operation > Processes > Click Details for process.
 - %SYS>do ^JOBEXAM.

	Job #	User	Namespace	Routine	Commands	Globals	State	CPU Time in ms	Elapsed Time	Process ID »	Parent PID	Device
-	1			CONTROL	0	0	RUN	9260	48:06:47	4504		
-	2			WRTDMN	5538	40354	RUNW	8080	48:06:47	4545		
-	3			GARCOL	0	0	RUNW	200	48:06:47	4546		
-	4			JRNDMN	28304	6858	RUNW	4800	48:06:47	4547		
-	6			AUXWD	2	0	RUNW	0	48:06:47	4548		
-	7			AUXWD	2	0	RUNW	0	48:06:47	4549		
-	8			AUXWD	2	0	RUNW	0	48:06:47	4550		
-	9			AUXWD	0	0	RUNW	0	48:06:47	4551		
-	10			AUXWD	0	0	RUNW	0	48:06:47	4552		
-	11			AUXWD	0	0	RUNW	0	48:06:47	4553		
-	12			AUXWD	0	0	RUNW	0	48:06:47	4554		
-	5			EXPDMN	0	0	RUNW	0	48:06:47	4555		
-	14		%SYS	MONITOR	142833	100	EVTW	1620	48:06:47	4557	4556	/dev/null
-	15		%SYS	CLNDMN	2869	291	RUNW	90	48:06:47	4558	4556	/dev/null
-	18		%SYS	LMFMON	228269	10639	EVTW	2140	48:06:47	4561	4556	/dev/null
Details	16		%SYS	RECEIVE	174842	16424	HANG	2330	48:06:47	4563	4556	/dev/null
-	20		%SYS	%SYS.SERVER	338	35	READ	40	48:06:47	4566	4565	[TCP]1972
Details	21		%SYS	%SYS.WorkQueueMgr	27847239	2925371	SEMW	70740	48:06:47	4569	4568	/dev/null
Details	22	TASKMGR	%SYS	%SYS.TaskSuper.1	595154	26463	EVTW	1810	48:06:47	4571	4568	/dev/null
-	23		%SYS	%SYS.Monitor.Control.1	15314307	1859211	EVTW	15340	48:06:47	4573	4568	/dev/null
Details	24	Daemon	%SYS	LOGDMN	3999535	95135	EVTW	17760	48:06:47	4574	4568	/dev/null
Details	25		%SYS	%SYS.WorkQueueMgr	987903	29952	SEMW	10530	48:06:47	4576	4568	/dev/null
Details	3	CSPSystem	%SYS	%SYS.cspServer3	12047	132	EVTW	930	48:06:44	4584	4566	[TCP]192.168



How To: Manage Processes in Portal

- System Operation > Processes > Click Details for process > Click action.

Process Details

ResumeSuspendTerminate

DetailsVariables

GENERAL INFORMATION	
Process ID:	13517
User Name:	irisowner
Login Roles:	%All
Escalated Roles:	None
OS User Name:	irisowner
Namespace:	USER

EXECUTION DETAILS	
Process State:	
In Transaction:	
Last Global Reference:	
Routine:	
Location:	
Source Location:	



How To: Manage Processes in ^JOBEXAM

- Actions available:
 - Terminate.
 - Suspend (and later Resume).
 - Examine.

IRIS for Windows (x86-64) 2018.1.1 (Build 611U)							
Job#	NSpace	Routine	Commands	Globals	State	PID	Current device
1		CONTROL	0	0	RUNW	5208	
2		WRTDMN	64	210	RUNW	3132	
3		GARCOL	0	0	RUNW	9204	
4		JRNDMN	1150	0	RUNW	2980	
5		EXPDMN	0	0	RUNW	5172	
6	%SYS	%SYS.cspServer2	312478	22281	READ	8568	localhost:51773
7	%SYS	MONITOR	8033	68	EVTW	11644	///./nul
8	%SYS	CLNDMN	195	19	RUNW	2496	///./nul
9	%SYS	RECEIVE	4046	220	HANGW	1040	///./nul
10	%SYS	ECPWork	0	0	RUNW	10868	///./nul
11	%SYS	%SYS.SERVER	0	0	READ	4208	51773:
12	%SYS	%SYS.cspServer2	455167	27661	READ	4848	localhost:51773
13	%SYS	LMFMON	190	18	HANGW	7500	///./nul
14	%SYS	%CSP.Daemon	5191	743	HANGW	9084	///./nul
15	%SYS	%SYS.ProcessQuery382115		5124	RUN	3832	TRM : 3832
16	%SYS	%SYS.cspServer3	2709	662	READ	10740	127.0.0.1:51773
17	%SYS	%SYS.Monitor.Con<435463		9156	EVTW	7984	///./nul
18	%SYS	%SYS.TaskSuper	15732	622	EVTW	8688	///./nul
19	%SYS	%SYS.cspServer2	6508	232	READ	4188	localhost:51773

(N)ext, (P)rev, (G)oto, (E)xamine, (T)erminate, (S)uspend, (R)esume, (Q)uit => █



How To: View Process Details

- System Operation > Processes > Click Details for process.
- In ^JOBEXAM routine, select "E" and provide PID #.

GENERAL INFORMATION	
Process ID:	4594
Parent PID:	4569
User Name:	
Login Roles:	%All
Escalated Roles:	None
OS User Name:	irisowner
Namespace:	%SYS
Process Priority:	2
Global References:	1289910
Private Global References:	146331
Commands Executed:	14101922
Memory Limit:	2147483646
Memory Peak:	2176
Memory Used:	690
Total CPU Time:	39850
Start Time:	2025-11-19 22:09:34
Elapsed Time:	48:16:51
Private Global Blocks:	9
Current Device:	/dev/null
Open Devices:	/dev/null*
Lock:	X, 1, ^["^"/Host/SYS/mgr/ irislocaldata/"]IRIS.WorkQueue("Worker",4594)



How To:

Terminate a Process

- System Operation > Processes > Click Details for process > Terminate.
 - If check box to send <RESJOB> error, creates entry in Application Error Log.
- %SYS>do ^JOBEXAM.
- %SYS>do ^RESJOB.
- **Not recommended to terminate process at OS level.**
 - Can result in ghost process.
 - Terminating System Process at OS level (WRTDMN, for example) causes fatal error.



Common Process States

EVT	waiting on event	RUN	examining job
HANG	executing HANG command	LOCK	waiting for lock
READ	reading from device	RUN	running
SEM	waiting on a semaphore	SUSP	suspended
WRT	writing to device	suffixes: H – halting, W – hibernating, D – dead	

- For more information on process states, refer to State property documentation in [%SYS.ProcessQuery class reference](#).



Process Locks

- Processes may lock data for concurrency control.
- To list and delete locks:
 - System Operation > Locks or
`%SYS>do ^LOCKTAB.`
- Displays:
 - Exclusive and shared locks.
 - Local and remote locks.



Managing Process Locks

- Actions:
 - Delete one lock.
 - Delete all of one job's locks.
 - ^LOCKTAB only:
 - Delete all of one computer's locks.
 - Delete all locks.
- Process exiting normally releases its locks.
 - Contact InterSystems Worldwide Response Center before deleting locks.
 - Typically better to terminate process than manually removing locks.
 - Removing locks can allow multiple processes to concurrently change same data.



Suggested Readings

- *System Administration Guide.*
 - *Managing InterSystems IRIS > Controlling InterSystems IRIS Processes.*



Summary

- What are the key points for this module?

